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Influence of intraoperative cerebral oximetry monitoring on neurocognitive function after coronary artery bypass surgery: a randomized, prospective study^{1†}

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Abstract

OBJECTIVES: Postoperative cognitive decline is common after cardiac surgery, but it is often unrecognized at the time of hospital discharge. However, it has a great impact on patient's quality of life. Cerebral oximetry with the INVOS (IN Vivo Optical Spectroscopy) system provides the possibility of non-invasive, continuous measurement of regional cerebral oxygen saturation (rSO₂), which can improve patients' outcome. The aim of this study was to examine whether cerebral oximetry can decrease the incidence of cognitive decline after coronary artery bypass grafting.

METHODS: We have performed a prospective, randomized study with 200 patients enrolled. Patients were divided into INVOS interventional group and CONTROL group without monitoring of cerebral oximetry. A standardized interventional protocol was performed in the INVOS group to maintain rSO₂ above 80% of the patient's baseline value or above 50% of the absolute value. Cognitive evaluation was performed in all patients before and 7 days after surgery. Logistic regression was used to reveal predictors of cognitive decline.

RESULTS: The incidence of cognitive decline 7 days after surgery was significantly lower ($P = 0.002$) in the INVOS interventional group (28%) than in the CONTROL group (52%). Intraoperative use of INVOS monitoring was associated with lower incidence of cognitive decline (odds ratio 0.21). In addition, predictors of cognitive decline revealed by multivariate logistic regression were older age, higher EuroSCORE and SAPS II (Simplified Acute Physiology Score) values, lower educational level and persistence of preoperative atrial fibrillation. Patients with prolonged rSO₂ desaturation, defined as rSO₂ area under the curve (AUC) of more than 150 min% for desaturation below 20% of baseline or AUC of more than 50 min% for desaturation below 50% absolute value, had an increased risk of cognitive decline.

CONCLUSION: Postoperative cognitive outcome was significantly better in patients with intraoperative cerebral oximetry monitoring. Prolonged rSO₂ desaturation is a predictor of cognitive decline and has to be avoided.

Keywords: Cerebral oximetry • Cardiopulmonary bypass • Cognitive dysfunction

INTRODUCTION

Cerebral injury following cardiac surgery is still frequent and has detrimental and often long-lasting consequences. It occurs in various forms, ranging from stroke, which is the most serious but relatively rare, to more frequent cognitive dysfunction (more than 50%), which may not be life threatening but has significant impact on the patient's quality of life following surgery [1, 2]. The aetiology of these complications is most likely multifactorial, involving embolism, hypoperfusion, hypoxia and inflammatory response,

which eventually lead to regional or global imbalance between cerebral oxygen demand and supply [3–9].

A large number of reports have revealed that the monitoring of regional cerebral oxygen saturation (rSO₂) with an INVOS (IN Vivo Optical Spectroscopy) system provides an opportunity for early recognition of this imbalance and subsequent performance of standardized interventions to prevent prolonged rSO₂ desaturation with the aim to avoid neurological and other major complications [10–13]. The INVOS system is based on transcranial near-infrared spectroscopy (NIRS), which provides continuous non-invasive measurement of rSO₂ in a small local area of cortex [14].

[†]Results of the study were presented at a meeting 'Hot topics in cardiac anesthesia and intensive care', Bol, Island of Brac, Croatia on 8 September 2012.

We present the results of the prospective, randomized study of 200 coronary artery bypass grafting (CABG) patients with the use of cardiopulmonary bypass (CPB). Patients were randomized into the interventional INVOS group, in which rSO_2 was maintained above 80% of patient's baseline value during the operation, and the CONTROL group, in which monitoring with the INVOS system was not used. The main objectives of the study were to examine whether the patients monitored with the INVOS system had lower incidence of cognitive dysfunction compared with those who were not monitored and to reveal predictive factors of cognitive decline after CABG surgery.

MATERIALS AND METHODS

Study population

The data were collected at the University Hospital Center Zagreb, and the study was approved by the hospital ethical committee. The study was preregistered in the ClinicalTrials.gov public database, and the registry number is NCT00917124. Two hundred patients between 40 and 80 years who underwent on-pump CABG surgery and signed informed consent were included in the study. These patients were recruited between June 2009 and September 2010. The study was designed as randomized, prospective and double blinded. Exclusion criteria were significant carotid artery stenosis, previous stroke or head injury, seizure, psychiatric illness, decompensated heart failure (NYHA III/IV), left ventricular ejection fraction less than 25%, emergency cardiac surgery, off-pump CABG and severely impaired renal and liver function.

Operation protocol

All patients received the same anaesthesia protocol, including premedication with intramuscular morphine 0.1 mg/kg and induction with midazolam, hypnomidate, sufentanyl and pancuronium. Anaesthesia was maintained with continuous sufentanyl infusion, propofol infusion (during CPB) and sevoflurane. Haemodynamic measurements were performed using the thermodilution method by a Swan Ganz catheter. All patients had standard intraoperative monitoring. Bispectral index or any other neurological monitoring were not used. After heparinization, mild hypothermic (32°C) CPB was established with flow rates of 2–2.5 l/min and haematocrit was maintained above 22%, mean arterial pressure over 60 mmHg and partial pressure of carbon dioxide between 4.5 and 6 kPa by alpha-stat management. There was no possibility to measure haematocrit values by optical sensor incorporated in CPB circuit during CPB. Therefore, they were obtained by analysing blood samples using a point-of-care analyser. Generally, a single-clamp technique was utilized. The rewarming rate during CPB was slow and it took approximately half an hour, without the occurrence of hyperthermia.

Randomization protocol

All patients were randomly assigned into the INVOS group or the CONTROL group using a computerized random number generator. After informed consent was obtained, an enclosed assignment in a sequentially numbered, opaque, sealed envelope was allocated to each patient. This envelope was opened by anaesthesiologist at the theatre, and the card inside determined whether the patient was in the INVOS or in the CONTROL group. The anaesthesiologists who

performed anaesthesia and interventional protocol for maintaining of cerebral oxygenation were involved neither in data collection nor in cognitive test results of patients. Investigators who performed cognitive tests were blinded to patient's allocation. Patients were also blinded to the allocation group. During surgery, patients involved in the INVOS group were monitored by cerebral oximetry using the INVOS system. Patients in the CONTROL group were not monitored with cerebral oximetry and only standardized monitoring in cardiac surgery was utilized. Figure 1 shows a flow diagram of patients through each stage of a randomized trial.

Cerebral oximetry

All patients in the INVOS group were monitored with the INVOS system (INVOS 5100C; Somanetics Corp., Troy, MI, USA). The INVOS system uses NIRS for non-invasive and continuous measurement of changes in cerebral oxygen saturation. The probes for INVOS cerebral oxygen monitoring were attached bilaterally on the patient's forehead overlying the frontal-temporal region to awaken patients who breathed oxygen by nasal catheter, just before induction of anaesthesia. A baseline regional cerebral oxygen saturation (rSO_2) value for each side of the brain was determined a short time after probes were attached. The rSO_2 values were displayed on a screen and recorded during the entire surgery. If the rSO_2 during surgery decreased below 80% of baseline value or below 50% of absolute value, we responded with standardized interventions to maintain rSO_2 above those values [14]. These interventions involved measures to eliminate mechanical obstruction to cerebral flow (repositioning of head or bypass cannulae), to increase cerebral oxygen delivery (increasing FiO_2 , pCO_2 , mean arterial pressure, cardiac output or pump flow and haematocrit) or to reduce cerebral oxygen consumption (increasing of anaesthetic depth and reduction of temperature). Despite those interventions, an inability of maintaining rSO_2 above the threshold of 20% under the baseline value was observed in some patients. Prolonged desaturation was determined as the rSO_2 area under the curve (AUC) of more than 150 min% under 20% of baseline value, or rSO_2 AUC of more than 50 min% under 50% of absolute value according to data from earlier studies [12, 13].

After surgery, AUCs were calculated for desaturations: more than 20% of baseline value (rSO_2 AUC < 20% of baseline value), more than 50% (rSO_2 AUC < 50% absolute value) and more than 40% of absolute value (rSO_2 AUC < 40% absolute value). AUCs account for both depth and duration of desaturation below these thresholds so they are expressed as min% values. If values of AUC from the left and right sides of the brain were different, we took the greater AUC value as relevant.

Neurocognitive assessment—primary outcome measure

Neurocognitive evaluation was performed preoperatively and 7 days after surgery using standardized tests [Mini-Mental State Examination (MMSE), Color Trail Test 1 (CTT 1) and Grooved-Pegboard Test (GP test)]. Cognitive impairment was defined as a decline in postoperative performance in one or more tests: decrease in an MMSE score for three or more points from baseline value and decrease of 1 SD (standard deviation) or more in performance on CTT 1 and GP tests. According to this definition, the cognitive outcome was assessed for each patient (dichotomous

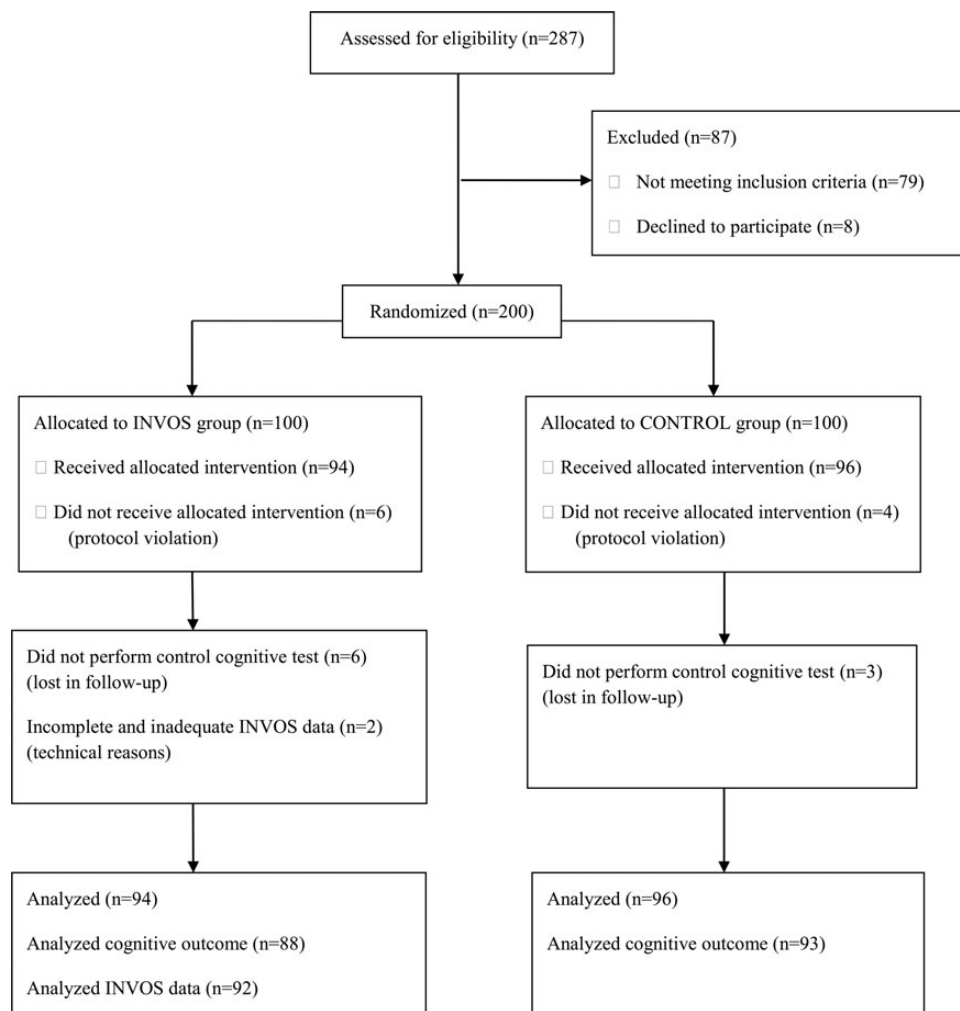


Figure 1: Flow diagram of patients allocated in the study.

outcome). Furthermore, all patients were divided into two groups according to the presence or absence of cognitive decline, and statistical analysis of preoperative, intraoperative and postoperative data was performed.

Secondary outcome measures

Patients were assessed for neurological deficit presented as coma, stupor, transient ischaemic attack (TIA) or stroke and for the second group of neuropsychological deficits presented as delirium. Coma is defined as a profound state of unconsciousness without response to verbal call, pain or any other stimulus. Stupor is defined as a state of unconsciousness from which the patient can be awakened only by vigorous physical stimulation. Stroke is characterized by the acute onset of a focal neurological deficit lasting more than 24 h. TIA is a clinical syndrome characterized by an acute loss of focal cerebral function lasting less than 24 h. Delirium or encephalopathy is characterized by confusion, agitation, disorientation and decreased alertness, but without an obvious focal neurological deficit.

Patients were also evaluated for major organ complications and for duration of intensive care unit (ICU) and overall hospital stay. Perioperative myocardial infarction was defined as an increase in troponin T values $>10 \times 99$ th percentile URL (upper reference limit) during the first 48 h following CABG, when associated with

the appearance of new pathological Q-waves or new left bundle branch blocks. All patients were on telemetry until hospital discharge and episodes of atrial fibrillation were observed. The duration of mechanical ventilation was defined as a period from the arrival in ICU until extubation. Prolonged mechanical ventilation was defined as the need for mechanical ventilation longer than 24 h. Infection was defined as clinically manifested and microbiologically confirmed infection.

A logistic value of EuroSCORE was calculated for all patients. SAPS II (Simplified Acute Physiology Score) is a severity of disease classification system for patients admitted to ICUs. SAPS II values were calculated at admission in ICU and at discharge from ICU. Carotid arteries Doppler ultrasound was routinely undertaken in all patients older than 65 years and in diabetic patients older than 60 years.

Statistical analysis

Sample size was determined assuming 50% incidence of cognitive impairment after cardiac surgery and the possibility of decreasing that incidence to 30% using cerebral oximetry. Based on 0.8 power to detect a significant difference ($P = 0.05$), 90 patients were required for each study group. We enrolled 100 patients per group due to the potential loss in the follow-up. All statistical

analyses were performed using Statistica 7.1 and SAS 8.2 software. Categorical data were analysed using the χ^2 test or Fisher's exact test, which is used when the expected frequency in any cell of a contingency table was below 5. According to normality of distribution, continuous variables were compared between groups using Student's *t*-test or non-parametric Mann-Whitney *U*-test as appropriate.

Multivariate logistic regression analysis with a stepwise procedure was used to define the risk factors for the development of cognitive decline. Two multivariate logistic models were made separately for preoperative and operative variables, listed as an appendix to Tables 3 and 4. The entrance level into multivariate logistic regression analysis was set to a *P*-value of less than or equal to 0.15. Univariate logistic regression analysis was also performed for all variables.

Continuous data are presented as mean $\pm$ SD. Categorical variables are presented as percentages. Statistical significance was considered at *P*-value of less than 0.05.

RESULTS

Figure 1 shows the flow diagram with information about enrolment, intervention allocation, follow-up and analysis of patients. There was good comparability between the two studied groups (Table 1). No statistically significant differences were observed between the INVOS and CONTROL group in patients' demographic and clinical characteristics and intraoperative and postoperative data. Nevertheless, there was a statistically significant difference between study groups in dichotomous cognitive outcome (*P* = 0.002)

Table 1: Comparison between the INVOS and CONTROL groups regarding demographic and clinical data

Variables	Group		P-value
	INVOS (n = 94)	Control (n = 96)	
Age (years)	61.9 $\pm$ 7.1	63.4 $\pm$ 8.8	0.177
Age categorical (% of patients ≥ 65 years)	32	50	0.017 ^a
Gender (female %)	20	24	0.655
EuroSCORE (%)	2.2 $\pm$ 1.7	2.4 $\pm$ 1.7	0.282
Left ventricular ejection fraction (%)	56 $\pm$ 9.7	56 $\pm$ 9.9	0.765
Years of education	11.3 $\pm$ 3.4	11.2 $\pm$ 3.2	0.811
Comorbidity (% of patients)			
Diabetes	30	34	0.602
Hypertension	84	88	0.367
Myocardial infarction within 1 month	9	13	0.370
Atrial fibrillation	4	5	0.757
Renal insufficiency	2	1	0.619
Number of grafts	2.5 $\pm$ 0.6	2.6 $\pm$ 0.6	0.640
Bypass time (min)	91 $\pm$ 31	89 $\pm$ 32	0.596
Aortic cross-clamp time (min)	63 $\pm$ 23	62 $\pm$ 26	0.669
Cardiac index before CPB (l/min/m ²)	2.3 $\pm$ 0.4	2.3 $\pm$ 0.5	0.722
Cardiac index after CPB (l/min/m ²)	2.9 $\pm$ 0.6	2.9 $\pm$ 0.6	0.928
'Single-clamp' technique (% of patients)	86	79	0.203
Erythrocytes concentrate during CPB (ml)	265 $\pm$ 306	181 $\pm$ 283	0.056
Erythrocytes concentrate after CPB (ml)	130 $\pm$ 212	214 $\pm$ 294	0.055
W/o Erythrocytes concentrate during surgery (% of patients)	36	36	0.967
W/o Erythrocytes concentrate during hospitalization (% of patients)	18	24	0.415
Inotropics (% of patients)	46	40	0.391
Erythrocytes concentrate in ICU (ml)	234 $\pm$ 333	239 $\pm$ 317	0.913
Cardiac index-ICU arrival (l/min/m ²)	2.9 $\pm$ 0.5	2.9 $\pm$ 0.6	0.461
Cardiac index-after 24 h (l/min/m ²)	3.1 $\pm$ 0.5	3.1 $\pm$ 0.5	0.648
Mechanical ventilation (h)	9.7 $\pm$ 4.2	9.3 $\pm$ 4.1	0.551
Postoperative complications (% of patients)			
Stroke, TIA, coma, stupor	4	1	0.204
Delirium	9	14	0.292
Cognitive impairment	28	52	0.002 ^a
Myocardial infarction	8	7	0.934
Atrial fibrillation	23	27	0.500
Prolonged mechanical ventilation	1	2	1.000
Haemodialysis	0	1	1.000
Infection	18	19	0.962
Revision	0	1	1.000
ICU length of stay (days)	2.7 $\pm$ 6.2	1.9 $\pm$ 0.9	0.443
Hospital stay > 7 days (% of patients)	49	45	0.729
SAPS II-arrival in ICU	18.9 $\pm$ 5.5	19.2 $\pm$ 5.5	0.655
SAPS II-ICU discharge	11.1 $\pm$ 3.6	11.4 $\pm$ 4.0	0.565

Data are expressed as frequencies % or $X \pm SD$ (mean $\pm$ standard deviation).

CPB: cardiopulmonary bypass; ICU: intensive care unit; SAPS: Simplified Acute Physiology Score; TIA: transient ischaemic attack; INVOS: IN Vivo Optical Spectroscopy.

^aStatistically significant value.

with lower incidence of cognitive decline in the INVOS group (INVOS 28 vs control 52%).

Data show a trend towards statistically significant difference between groups regarding blood transfusion, with higher use of blood during CPB observed in the INVOS group ($P = 0.056$). In the CONTROL group, in contrast, there was a trend towards statistically significant higher use of blood after CPB ($P = 0.055$). The percentage of patients who did not receive a blood transfusion during surgery, and the percentage of those who did not receive any blood during hospital stay, was without a statistically significant difference between study groups (Table 1).

All patients with cognitive decline (40.3%) were compared with those without decline, and a large number of variables show statistically significant differences between these two groups (Table 2).

The results from multivariate logistic regression of all patients revealed that intraoperative use of INVOS monitoring was

associated with lower incidence of cognitive decline [odds ratio (OR) 0.21] (Table 3). In addition, predictors of postoperative cognitive decline were older age, lower educational level, higher EuroSCORE value, preoperative persistence of atrial fibrillation and higher value of SAPS II at the moment of ICU discharge. The results from univariate logistic regression analysis are also presented in Table 3. Multivariate analysis within the INVOS group revealed prolonged rSO_2 desaturation during surgery as an important risk factor for cognitive decline (Table 4).

DISCUSSION

In the present study, the overall incidence of early postoperative cognitive impairment was 40.3%, defined as a decrease in performance on at least one of the cognitive tests used. We found a statistically significant difference ($P = 0.002$) in the incidence of

Table 2: Comparison between patients with and without cognitive decline

Variables	Without cognitive decline (n = 108)	With cognitive decline (n = 73)	P-value
Age (years)	59.6 ± 7.4	66.9 ± 6.9	<0.001 ^a
Gender (female %)	26	21	0.903
EuroSCORE (%)	1.9 ± 1.4	2.9 ± 1.9	<0.001 ^a
Left ventricular ejection fraction (%)	56.1 ± 9.5	56.2 ± 9.8	0.918
Years of education	11.8 ± 2.9	10.6 ± 3.7	0.033 ^a
Comorbidity (% of patients)			
Diabetes	30	33	0.764
Hypertension	84	89	0.360
Myocardial infarction within 1 month	7	8	0.070
Atrial fibrillation	1	10	0.016 ^a
Renal insufficiency	0	1	0.403
Number of grafts	2.5 ± 0.6	2.6 ± 0.6	0.389
Bypass time (min)	86 ± 29	93 ± 32	0.156
Aortic cross-clamp time (min)	61 ± 22	63 ± 27	0.519
Cardiac index before CPB (l/min/m ²)	2.42 ± 0.5	2.17 ± 0.5	<0.001 ^a
Cardiac index after CPB (l/min/m ²)	3.00 ± 0.6	2.81 ± 0.5	0.036 ^a
'Single-clamp' technique (% of patients)	87%	74%	0.025 ^a
Erythrocytes concentrate during CPB (ml)	202 ± 265	247 ± 333	0.556
Erythrocytes concentrate after CPB (ml)	119 ± 211	242 ± 302	0.002 ^a
W/o Erythrocytes concentrate during surgery (% of patients)	48	29	0.022 ^a
W/o Erythrocytes concentrate during hospitalization (% of patients)	28	13	0.027 ^a
Inotropics (% of patients)	33	52	0.012 ^a
Erythrocytes concentrate in ICU (ml)	196 ± 314	291 ± 340	0.056
Cardiac index-ICU arrival (l/min/m ²)	3.0 ± 0.6	2.8 ± 0.6	0.016 ^a
Cardiac index-after 24 h (l/min/m ²)	3.1 ± 0.5	3.1 ± 0.5	0.648
Mechanical ventilation (h)	8.6 ± 3.4	10.4 ± 4.4	0.004 ^a
Postoperative complications (% of patients)			
Stroke, TIA, coma, stupor	0	4	0.065
Delirium	5	18	0.005 ^a
Myocardial infarction	7	8	0.669
Atrial fibrillation	23	26	0.683
Prolonged mechanical ventilation	0	1	0.406
Haemodialysis	0	1	1.000
Infection	15	22	0.230
Revision	1	0	1.000
ICU length of stay (days)	1.8 ± 1.0	2.1 ± 0.9	0.037 ^a
Hospital stay > 7 days (% of patients)	41	55	0.107
SAPS II (arrival in ICU)	17.3 ± 4.6	21.0 ± 4.9	<0.001 ^a
SAPS II (ICU discharge)	9.9 ± 3.3	12.9 ± 3.5	<0.001 ^a

Data are expressed as frequencies % or $X \pm SD$ (mean ± standard deviation).

CPB: cardiopulmonary bypass; ICU: intensive care unit; SAPS: Simplified Acute Physiology Score.

^aStatistically significant value.

Table 3: Predictors of early postoperative cognitive decline

Variables	Multivariate logistic regression—all patients OR (95% CI)	Univariate logistic regression—all patients OR (95% CI)
Age	1.15 (1.09–1.22) ^a	1.15 (1.09–1.21)
EuroSCORE	1.35 (1.09–1.67) ^a	1.42 (1.15–1.74)
Years of education	0.87 (0.79–0.97) ^a	0.89 (0.81–0.98)
Atrial fibrillation—preoperatively	13.16 (1.5–115.5) ^a	11.34 (1.37–94.2)
Cardiac index before CPB	–	0.28 (0.13–0.6)
Use of INVOS monitoring	0.21 (0.07–0.57) ^b	0.37 (0.20–0.69)
Erythrocytes concentrate during CPB	–	1.053 (1.052–1.054)
Erythrocytes concentrate after CPB	–	1.208 (1.207–1.209)
SAPS II—arrival in ICU	–	1.17 (1.09–1.26)
SAPS II—ICU discharge	1.38 (1.18–1.60) ^b	1.29 (1.17–1.42)

Univariate logistic regression analysis was performed for all listed variables and only statistically significant results are presented in the table.

OR: odds ratio; 95% CI: 95% confidence interval; CPB: cardiopulmonary bypass; ICU: intensive care unit; SAPS: Simplified Acute Physiology Score.

^aIn the first multivariate logistic regression model—preoperative—independent variables were age, gender, years of education, diabetes, hypertension, preoperative atrial fibrillation, myocardial infarction within 1 month before surgery, beta blockers, ACE inhibitors, Ca-channel blockers, antilipemics, EuroSCORE and left ventricular ejection fraction.

^bIn the second multivariate logistic regression model—operative—independent variables were number of grafts, CPB time, aortic cross-clamp time, single-clamp technique, use of INVOS monitoring, CI before and after CPB, CI after admission in ICU and 24 h after surgery, volume of transfused erythrocyte concentrate—during CPB, after CPB and in ICU, troponin values after admission in ICU, on the first and the second postoperative day, time period until extubation, prolonged mechanical ventilation, delirium, postoperative infection, neurological injury—stroke, TIA, comma and stupor, postoperative atrial fibrillation, postoperative myocardial infarction, ICU stay, SAPS II at admission in ICU and at discharge from ICU.

Table 4: Predictors of postoperative cognitive decline—INVOS group

Variables	Multivariate logistic regression—INVOS group OR (95% CI)	Univariate logistic regression—INVOS group OR (95% CI)
Age	1.12 (1.03–1.21) ^a	1.12 (1.04–1.20)
Years of education	–	0.83 (0.71–0.96)
Erythrocytes concentrate during CPB	–	1.129 (1.127–1.13)
Erythrocytes concentrate after CPB	–	1.081 (1.080–1.081)
SAPS II—arrival in ICU	1.33 (1.07–1.67) ^b	1.17 (1.04–1.31)
SAPS II—ICU discharge	–	1.24 (1.06–1.46)
Prolonged rSO ₂ desaturation	12.1 (2.4–60.7) ^b	5.0 (1.85–13.5)
Nadir rSO ₂ during operation	–	0.93 (0.87–0.99)
rSO ₂ AUC < 20% baseline value	–	1.026 (1.025–1.028)
rSO ₂ AUC < 50% absolute value	–	1.037 (1.034–1.041)

Univariate logistic regression analysis was performed for all listed variables and only statistically significant results are presented in the table.

Prolonged rSO₂ desaturation = rSO₂ AUC < 20% baseline of more than 150 min or rSO₂ AUC < 50% absolute of more than 50 min.

OR: odds ratio; 95% CI: 95% confidence interval; CPB: cardiopulmonary bypass; SAPS: Simplified Acute Physiology Score; ICU: intensive care unit; rSO₂: regional cerebral oxygen saturation; AUC: area under the curve.

^aIn the first multivariate logistic regression model—preoperative—independent variables were age, sex, years of education, diabetes, hypertension, preoperative atrial fibrillation, myocardial infarction within 1 month before surgery, beta blockers, ACE inhibitors, Ca-channel blockers, antilipemics, EuroSCORE and ejection fraction.

^bIn the second multivariate logistic regression model—operative—independent variables were number of grafts, CPB time, aortic cross-clamp time, single-clamp technique, CI before and after CPB, CI after admission in ICU and 24 h after surgery, volume of transfused erythrocyte concentrate—during CPB, after CPB and in ICU, troponin values after admission in ICU, on the first and the second postoperative day, time period until extubation, prolonged mechanical ventilation, delirium, postoperative infection, neurological injury—stroke, TIA, comma and stupor, postoperative atrial fibrillation, postoperative myocardial infarction, ICU stay, SAPS II at admission in ICU and at discharge from ICU, prolonged rSO₂ desaturation, nadir rSO₂ during operation, rSO₂ AUC < 20% of baseline value, rSO₂ AUC < 50% absolute value, rSO₂ AUC < 40% absolute value.

cognitive impairment between the INVOS (28%) and CONTROL group (52%). In addition, the results of multivariate logistic regression show that the use of INVOS monitoring has a predictive value in terms of lower incidence of early postoperative cognitive decline.

The first published prospective randomized study performed to examine whether the patients monitored with the INVOS system

had lower incidence of cognitive dysfunction was performed by Slater *et al.* They monitored 240 CABG patients with the INVOS system in both control (blind monitoring) and interventional groups. Early postoperative cognitive decline developed in 60% of 240 patients, but no statistically significant difference was found in the incidence of cognitive decline between the interventional (58%) and control (61%) groups, as assessed with a battery of

standardized neuropsychological tests [13]. Prolonged rSO₂ desaturation was revealed as a strong predictor of cognitive decline, but the authors observed almost the same incidence of prolonged rSO₂ desaturation in the interventional group (26%) as in the CONTROL group (30%), which might have resulted in no difference in cognitive decline being found between the groups. The observed lack of impact on duration of rSO₂ desaturation in the INVOS group intervention was explained as due to poor compliance with the treatment protocol. On the contrary, in the circumstances of our study where blind monitoring of cerebral oximetry in the CONTROL group was not provided, we can only presume that the observed higher incidence of cognitive decline in the CONTROL group was associated with possible greater prolongation of rSO₂ desaturation. Furthermore, although statistically significant differences between groups in regard to age were not found (continuous data), categorical data show that there were significantly more patients ($n = 48$) in the age range of ≥ 65 years in the CONTROL group compared with the INVOS group ($n = 30$), which could have also had an impact on higher incidence of cognitive decline in the CONTROL group.

Predictors of early postoperative cognitive decline

Patients with cognitive decline in our study were significantly older than those without it, and age was revealed as a predictor of cognitive decline as well. That could be explained by the possible presence of aortic atherosclerosis and unrecognized cerebrovascular disease in older patients, followed by aberrations in cerebral autoregulation. Earlier studies also defined age as a predictor of cognitive decline in CABG patients [15, 16].

Persistence of preoperative atrial fibrillation was identified as a strong predictor of cognitive decline, but there were only a small number of those patients in our study. Atrial fibrillation is a possible source of thromboembolism and it is also associated with decreased cardiac index, both of which can cause an increased risk for cognitive decline. Previous studies reported the impact of postoperative atrial fibrillation on neurocognitive outcome after CABG, but according to results of logistic regression in our study, we did not determine that impact [17, 18].

In the present study, lower educational level was associated with cognitive decline, as was observed in previously conducted similar studies. The mechanism of this observation is unknown, but protective effects of higher education in the development of Alzheimer's disease were also previously described in the literature [19]. It was observed that less educated patients were a bit older, but it could also be speculated that the rate of hypertension, hypercholesterolaemia and obesity is higher in less educated patients regardless of age, which all may influence the higher incidence of cognitive decline in that group.

Cardiac index values before and after CPB and also after surgery were significantly lower in the group of patients with cognitive decline, and consequently they needed a greater amount of inotropic support. These results suggest that lower cardiac function is associated with impairment in cognitive function.

In our study, patients with early postoperative cognitive decline received significantly higher volumes of erythrocyte concentrate transfusion after CPB in comparison with patients without cognitive decline. Moreover, higher volumes of erythrocyte transfusion was associated with the development of cognitive decline. In addition, there were significantly more patients who did not receive any transfusion during hospital stay in the group of

patients without postoperative cognitive decline compared with the group with postoperative cognitive decline ($P = 0.02$). Although many studies have pointed out the drawbacks of low values of haematocrit during CPB, there is still no precise haematocrit value at which benefit of transfusion overcomes its risks [20–22]. In the present study, the INVOS group received almost the same volume of erythrocyte concentrate as the CONTROL group, so the risk of receiving a blood transfusion was not increased as a consequence of monitoring rSO₂. Nevertheless, we observed that more erythrocyte concentrate in the INVOS group was administered during CPB in contrast to the CONTROL group where the majority was given after CPB. Namely, cerebral oxygen supply/demand imbalance caused by anaemia was early recognized by rSO₂ desaturation. Therefore, erythrocyte concentrate was given earlier in the INVOS group than in the CONTROL group, which may have resulted in better cognitive outcome observed in the INVOS group. The blood transfusion was assumed as intervention only in cases when cerebral desaturation was followed by a decrease in haematocrit value to less than 22%.

Prolonged rSO₂ desaturation

Prolonged rSO₂ desaturation, defined as a value of rSO₂ AUC<20% basal value greater than 150 min% or a value of rSO₂ AUC<50% absolute value greater than 50 min%, was observed in 38% of patients in the INVOS group. Namely, interventions that were undertaken during monitoring of brain rSO₂ in the majority of patients resulted in recovery of decreased rSO₂ to baseline values. However, standardized interventions were not always effective in some patients and prolonged rSO₂ desaturation occurred. The number of interventions undertaken to maintain rSO₂ above 80% of baseline value in the INVOS group was 8, on average.

Multiple logistic regression analysis of our data revealed prolonged rSO₂ desaturation as the strongest predictor of cognitive decline in the INVOS group (OR 12.1; CI 2.4–60.7; $P = 0.003$).

Univariate logistic regression also showed a predictive value of the nadir rSO₂ during surgery—a lower nadir rSO₂ was associated with a higher possibility of cognitive decline and there were also predictive values of rSO₂ AUC<20% baseline value and rSO₂ AUC<50% absolute value. Previous studies also revealed that prolonged rSO₂ desaturation was associated with early postoperative cognitive decline [11, 13].

Limitations of the study

The major possible limitation of this study is the number and selection of performed neurocognitive tests. We used CTT and GPT, which were recommended by the 'Statement of consensus on assessment of neurobehavioral outcomes after cardiac surgery' [23]. An MMSE test is not part of the recommended core neuropsychological battery because of its inadequate reliability, but we used it since it provides a quick orientation in several domains of patients' cognition. Nevertheless, the selected tests were easy to perform, and patients' motivation and attention were simply maintained during the study. The second important limitation is the lack of blind monitoring of cerebral oximetry in the CONTROL group, which would have given information on incidence of prolonged desaturation in this group. In such a way, we would have been able to demonstrate more clearly the impact of rSO₂ desaturation treatment on better cognitive outcome in the

interventional group. Last, but not least, the determination of aortic atheromatous disease grade in the preoperative period was not performed due to limitations in equipment capacity.

CONCLUSION

Intraoperative cerebral oximetry monitoring and interventions performed to correct cerebral rSO₂ desaturation have resulted in a better cognitive outcome of CABG patients under conditions of our study. Prolonged cerebral desaturation during operations is revealed as an important predictor of cognitive decline and should be avoided.

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